

### Product Summary

|                   |                                 |
|-------------------|---------------------------------|
| $V_{CE}$          | 650 V                           |
| $I_C$             | 75A @ $T_C=100^{\circ}\text{C}$ |
| $V_{CE(sat),Typ}$ | 1.6V @ $I_C=75\text{A}$         |

### Trench Field Stop IGBT Co-packed with SiC Schottky Barrier Diode

#### Features

- Low  $V_{CE(sat)}$
- Trench FS Technology
- High Speed Switching
- Hybrid SiC Discrete Devices
- Halogen Free, RoHS Compliant

#### Applications

- UPS
- PV Inverter
- Welding Machine
- On-board Charger
- Totem Pole Bridgeless PFC
- High Frequency Motor Drive

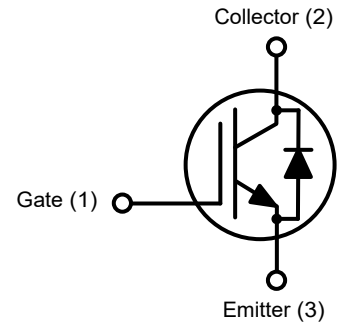
#### Package Pin Definitions

- Pin1 - Gate
- Pin2 - Collector & Backside
- Pin3 - Emitter

#### Package Parameters

| Part Number | Marking     | Package  |
|-------------|-------------|----------|
| BGH75N65HF1 | BGH75N65HF1 | TO-247-3 |

### Package: TO-247-3



**Maximum Ratings( $T_c=25^\circ\text{C}$  Unless Otherwise Noted)**

| Symbol        | Parameter                                              |                                                   | Value          | Unit             |
|---------------|--------------------------------------------------------|---------------------------------------------------|----------------|------------------|
| $V_{CE}$      | Collector-Emitter Voltage, $T_j \geq 25^\circ\text{C}$ |                                                   | 650            | V                |
| $V_{GE}$      | Gate-Emitter Voltage                                   |                                                   | $\pm 20$       |                  |
|               | Transient Gate-Emitter Voltage                         |                                                   | $\pm 30$       |                  |
| $I_C$         | DC Collector Current, limited by $T_{jmax}$            | $T_c=25^\circ\text{C}$                            | 135            | A                |
|               |                                                        | $T_c=100^\circ\text{C}$                           | 75             |                  |
| $I_F$         | Diode Forward Current, limited by $T_{jmax}$           | $T_c=25^\circ\text{C}$                            | 70             | A                |
|               |                                                        | $T_c=100^\circ\text{C}$                           | 47             |                  |
| $I_{C,pulse}$ | Pulse Collector Current                                | $V_{GE}=15\text{V}$ , $t_p$ limited by $T_{jmax}$ | 300            | A                |
| $P_{tot}$     | Power Dissipation                                      | $T_c=25^\circ\text{C}$                            | 338            | W                |
| $T_j$         | Operating Junction Temperature                         |                                                   | $-40 \sim 150$ | $^\circ\text{C}$ |
| $T_{stg}$     | Storage Temperature Range                              |                                                   | $-55 \sim 150$ | $^\circ\text{C}$ |
| $M_d$         | TO-247 mounting torque                                 | M3 Screw                                          | 0.7            | Nm               |

**Thermal Resistance**

| Symbol        | Parameter                                 | Value |      |      | Unit |
|---------------|-------------------------------------------|-------|------|------|------|
|               |                                           | Min.  | Typ. | Max. |      |
| $R_{th(j-c)}$ | IGBT Thermal Resistance-Junction to Case  |       | 0.37 |      | K/W  |
| $R_{th(j-c)}$ | Diode Thermal Resistance-Junction to Case |       | 0.57 |      | K/W  |
| $R_{th(j-a)}$ | Thermal Resistance-Junction to Ambient    |       | 30   |      | K/W  |

**Electrical Characteristics (Defined at  $T_j=25^\circ\text{C}$  Unless Otherwise Specified)**
**IGBT Static Characteristics**

| Symbol        | Parameter                            | Test Conditions                              |                         | Min. | Typ. | Max. | Unit          |
|---------------|--------------------------------------|----------------------------------------------|-------------------------|------|------|------|---------------|
| $V_{CE(sat)}$ | Collector-Emitter Saturation Voltage | $V_{GE}=15\text{V}$ , $I_C=75\text{A}$       | $T_j=25^\circ\text{C}$  |      | 1.6  | 2.1  | V             |
|               |                                      |                                              | $T_j=100^\circ\text{C}$ |      | 1.79 |      |               |
|               |                                      |                                              | $T_j=150^\circ\text{C}$ |      | 1.94 |      |               |
| $I_{CES}$     | Zero Gate Voltage Drain Current      | $V_{CE}=650\text{V}$ , $V_{GE}=0\text{V}$    | $T_j=25^\circ\text{C}$  |      |      | 500  | $\mu\text{A}$ |
|               |                                      |                                              | $T_j=150^\circ\text{C}$ |      |      | 2000 |               |
|               |                                      | $V_{CE}=480\text{V}$ , $V_{GE}=0\text{V}$    | $T_j=25^\circ\text{C}$  |      |      | 90   |               |
| $I_{GES}$     | Gate-Emitter Leakage Current         | $V_{GE}=\pm 20\text{V}$ , $V_{CE}=0\text{V}$ | $T_j=25^\circ\text{C}$  |      |      | 100  | nA            |

|              |                        |                               |                  |     |    |     |   |
|--------------|------------------------|-------------------------------|------------------|-----|----|-----|---|
| $V_{GE(th)}$ | Gate Threshold Voltage | $V_{GE}=V_{CE}, I_C=750\mu A$ | $T_J=25^\circ C$ | 4.2 | 5  | 5.8 | V |
| $g_{fs}$     | Transconductance       | $V_{CE}=20V, I_C=75A$         |                  |     | 94 |     | S |

### Dynamic Characteristics

| Symbol    | Parameter                    | Test conditions                       | Value |      |      | Unit |
|-----------|------------------------------|---------------------------------------|-------|------|------|------|
|           |                              |                                       | Min.  | Typ. | Max. |      |
| $C_{ies}$ | Input Capacitance            | $V_{GE}=0V, V_{CE}=25V$<br>$f=250kHz$ |       | 8164 |      | pF   |
| $C_{oes}$ | Output Capacitance           |                                       |       | 468  |      | pF   |
| $C_{res}$ | Reverse Transfer Capacitance |                                       |       | 137  |      | pF   |
| $Q_G$     | Gate Charge                  | $V_{CC}=520V, V_{GE}=15V, I_C=75A$    |       | 444  |      | nC   |

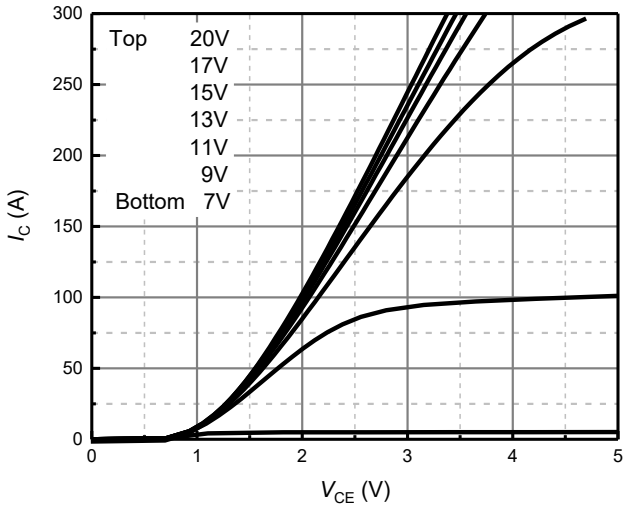
### Switching Characteristics, Inductive Load

| Symbol       | Parameter              | Test conditions                                                                                  | Value |      |      | Unit |
|--------------|------------------------|--------------------------------------------------------------------------------------------------|-------|------|------|------|
|              |                        |                                                                                                  | Min.  | Typ. | Max. |      |
| $t_{d(on)}$  | Turn-On Delay Time     | $T_J=25^\circ C$<br>$V_{DC}=400V, I_C=75A$<br>$V_{GE}=0/15V, R_{G(ext)}=4.7\Omega$<br>$L_o=60nH$ |       | 27   |      | ns   |
| $t_r$        | Rise Time              |                                                                                                  |       | 78   |      |      |
| $t_{d(off)}$ | Turn-Off Delay Time    |                                                                                                  |       | 239  |      |      |
| $t_f$        | Fall Time              |                                                                                                  |       | 59   |      |      |
| $E_{on}$     | Turn-On Energy         |                                                                                                  |       | 2044 |      | uJ   |
| $E_{off}$    | Turn-Off Energy        |                                                                                                  |       | 1192 |      |      |
| $E_{total}$  | Total Switching Energy |                                                                                                  |       | 3236 |      |      |

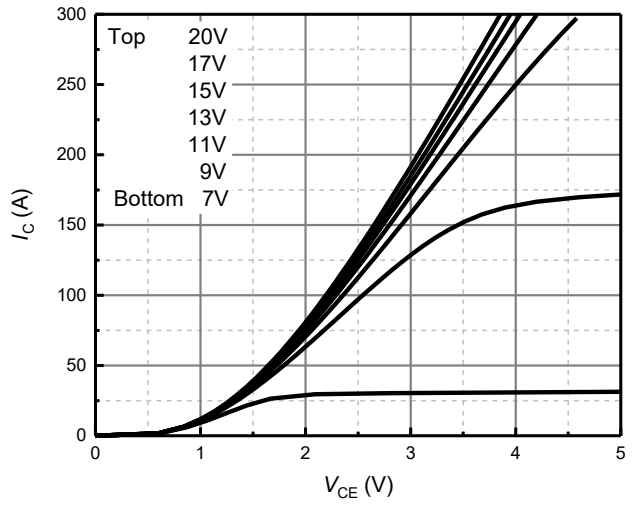
### SiC Schottky Barrier Diode Static Characteristics

| Symbol | Parameter               | Test conditions            |                   | Value |      | Unit |
|--------|-------------------------|----------------------------|-------------------|-------|------|------|
|        |                         |                            | Min.              | Typ.  | Max. |      |
| $V_F$  | Diode Forward Voltage   | $I_F=30A, V_{GE}=0V$       | $T_J=25^\circ C$  | 1.51  |      | V    |
|        |                         |                            | $T_J=100^\circ C$ | 1.63  |      |      |
|        |                         |                            | $T_J=150^\circ C$ | 1.81  |      |      |
| $Q_C$  | Diode Capacitive Charge | $V_R=400V, T_J=25^\circ C$ |                   | 64    |      | nC   |
| $C$    | Diode Capacitance       | $V_R=1V, f=1MHz$           |                   | 998   |      | pF   |
|        |                         | $V_R=300V, f=1MHz$         |                   | 110   |      |      |

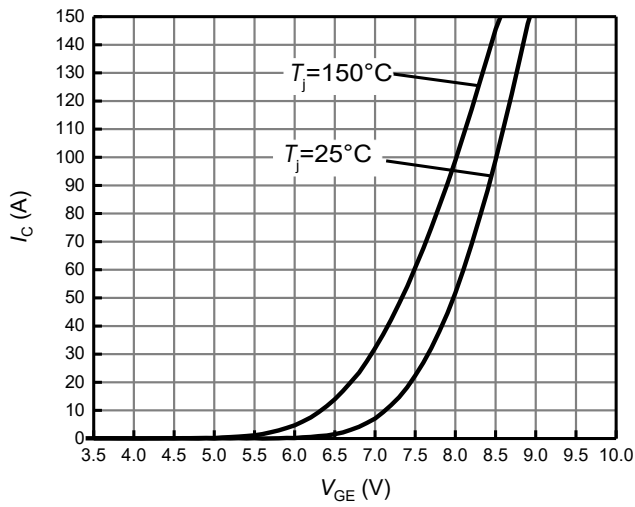
### Typical Performance



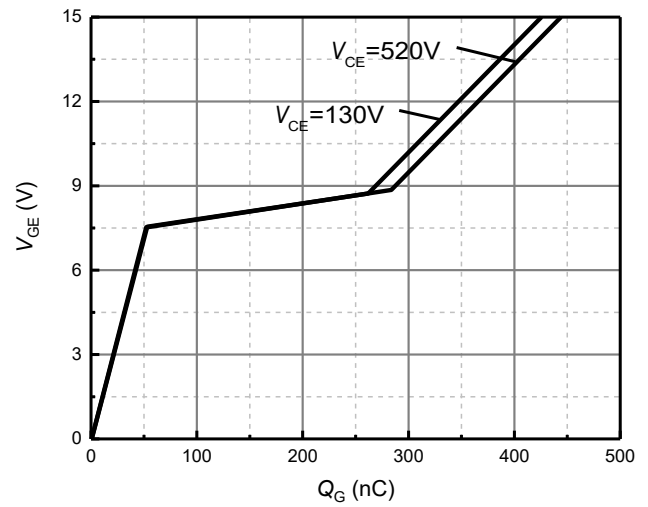
**Figure 1** Output Characteristics  
( $T_j = 25^\circ\text{C}$ )



**Figure 2** Output Characteristics  
( $T_j = 150^\circ\text{C}$ )

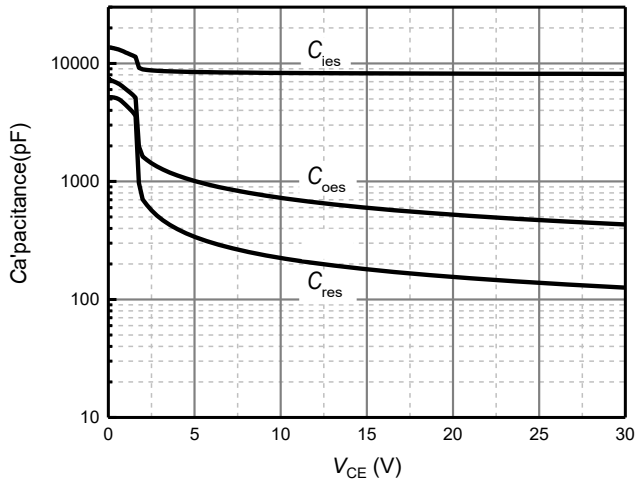


**Figure 3** Transfer Characteristics for Various Temperature  
( $V_{CE} = 20\text{V}$ )

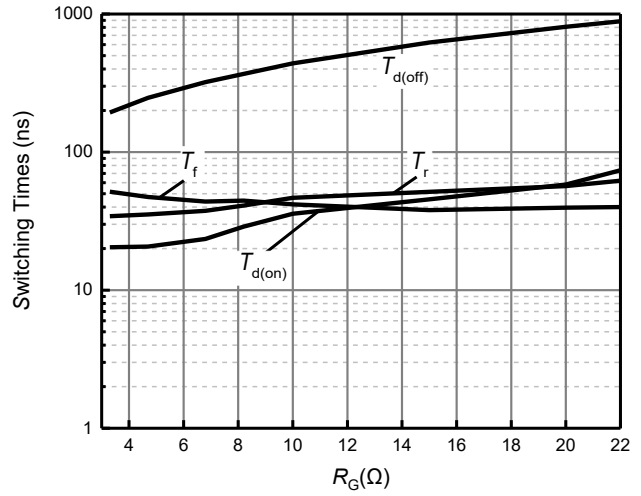


**Figure 4** Gate Charge Characteristics  
( $I_C = 75\text{A}$ )

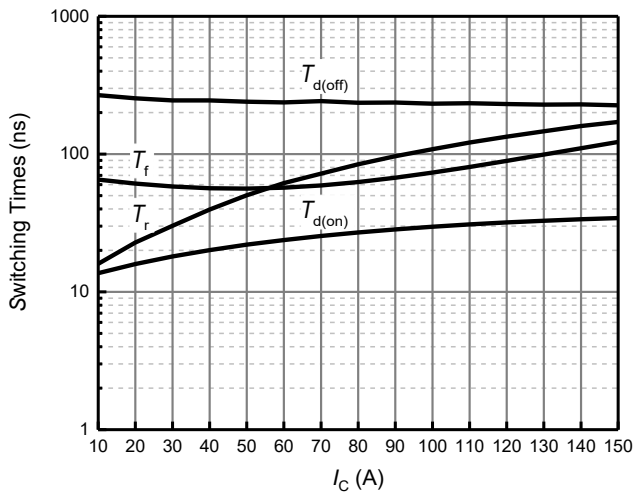
## Typical Performance



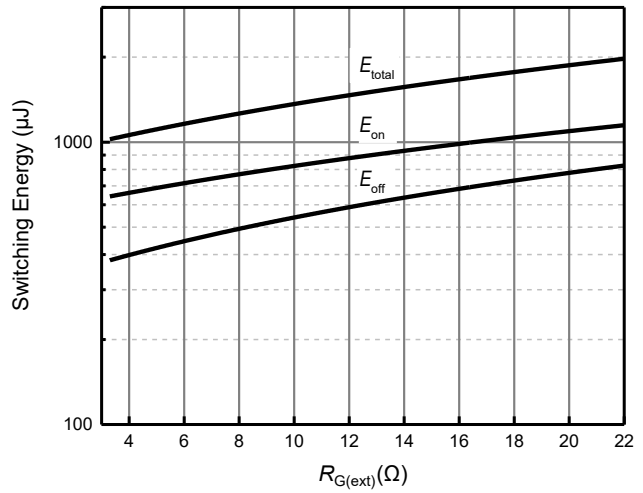
**Figure 5** Capacitance Characteristics  
( $V_{GE}=0V$ ,  $f=250kHz$ )



**Figure 6** Switching Times vs. Gate Resistor  
( $V_{DC}=400V$ ,  $V_{GE}=0/15V$ ,  $I_C=37.5A$ ,  $T_j=25^\circ C$ )

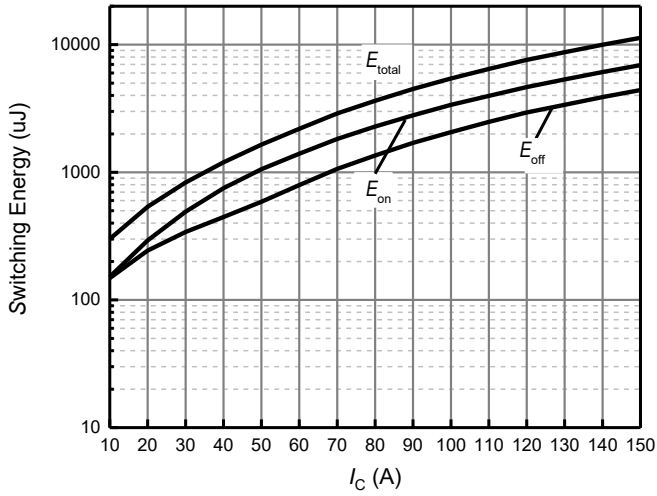


**Figure 7** Switching Times vs. Collector Current  
( $V_{DC}=400V$ ,  $V_{GE}=0/15V$ ,  $R_{G(ext)}=4.7\Omega$ ,  $T_j=25^\circ C$ )

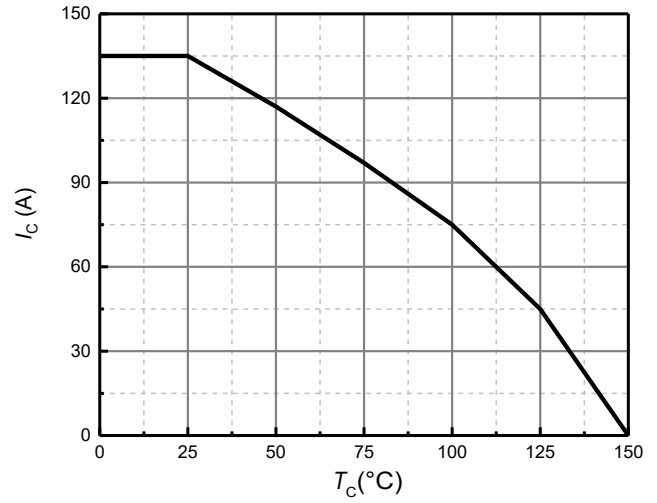


**Figure 8** Switching Loss vs. Gate Resistor  
( $V_{DC}=400V$ ,  $V_{GE}=0/15V$ ,  $I_C=37.5A$ ,  $T_j=25^\circ C$ )

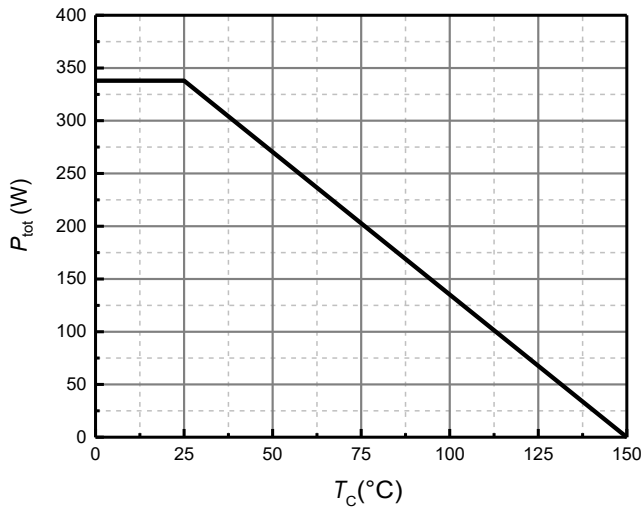
## Typical Performance



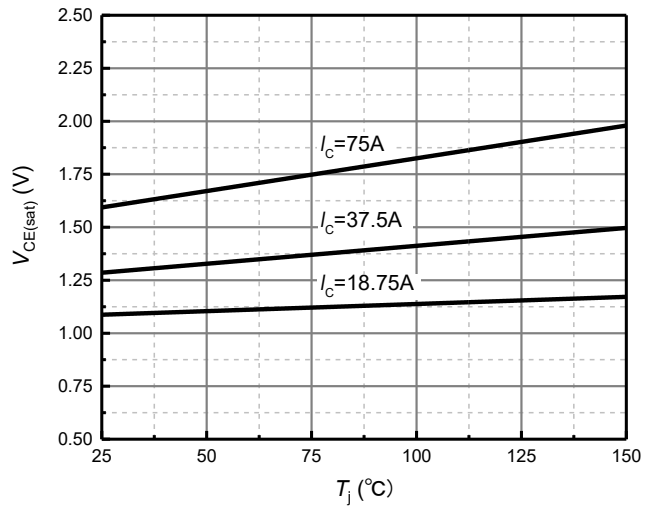
**Figure 9** Switching Loss vs. Collector Current ( $V_{DC}=400V$ ,  $V_{GE}=0/15V$ ,  $R_{G(ext)}=4.7\Omega$ ,  $T_J=25^\circ C$ )



**Figure 10** Collector Current vs. Case Temperature

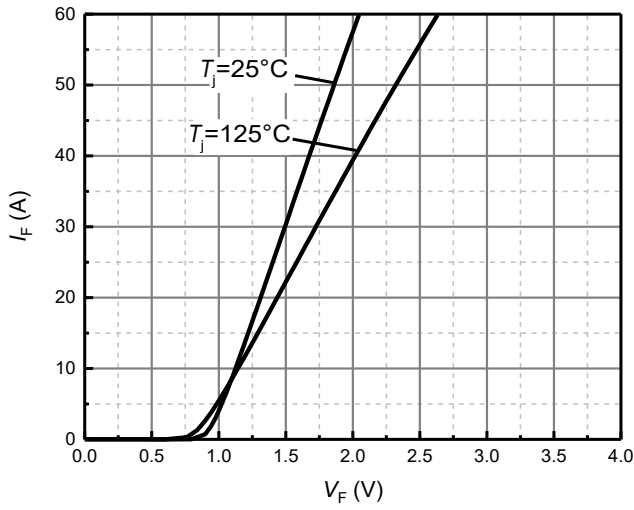


**Figure 11** Power Dissipation vs. Case Temperature

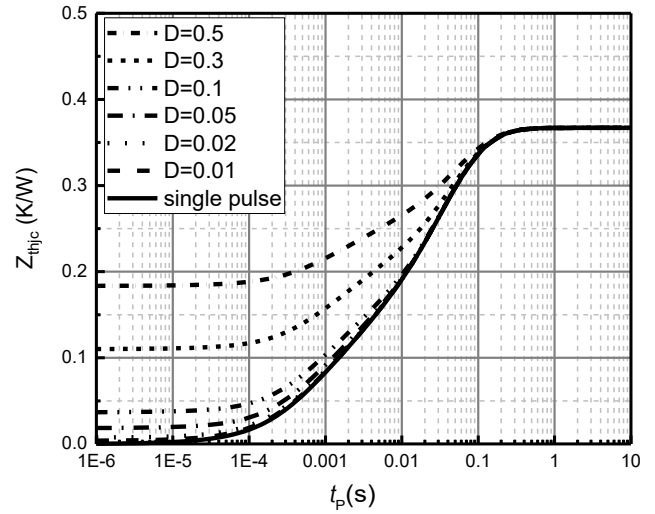


**Figure 12** Collector-Emitter Saturation Voltage vs. Junction Temperature for Various Collector Current ( $V_{GE}=15V$ )

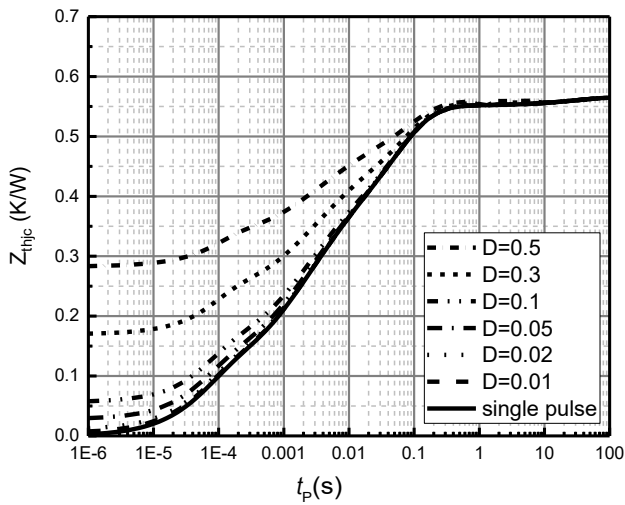
### Typical Performance



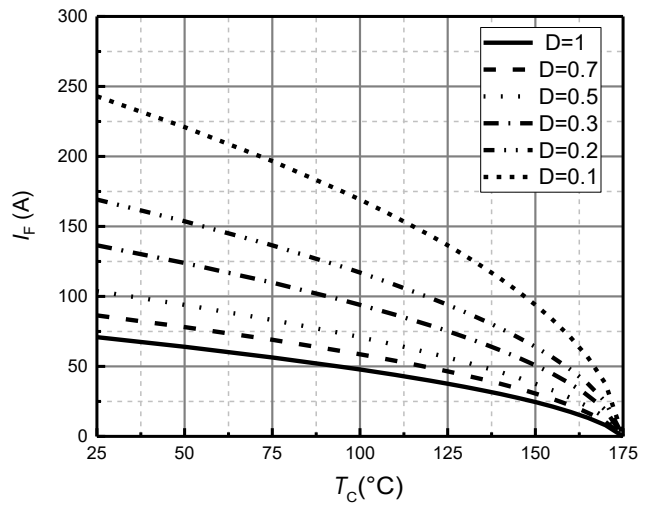
**Figure 13 Forward Characteristic of Diode**



**Figure 14 IGBT Transient Thermal Response Curve**

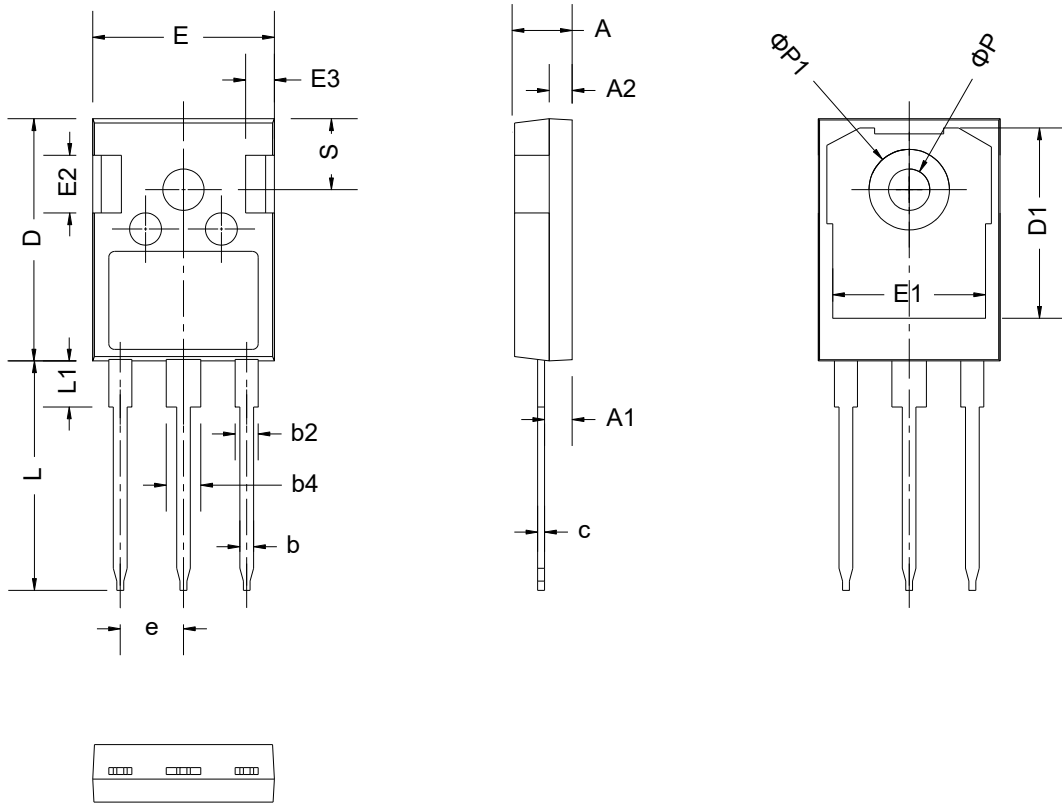


**Figure 15 Diode Transient Thermal Response Curve**



**Figure 16 Diode Forward Current as Function of Temperature, D=duty cycle**

## Package Dimensions



| Items | mm       |       |       |
|-------|----------|-------|-------|
|       | MIN      | NOM   | MAX   |
| A     | 4.80     | 5.00  | 5.20  |
| A1    | 2.21     | 2.41  | 2.59  |
| A2    | 1.85     | 2.00  | 2.15  |
| b     | 1.11     | 1.21  | 1.36  |
| b2    | 1.91     | 2.01  | 2.21  |
| b4    | 2.91     | 3.01  | 3.21  |
| c     | 0.51     | 0.61  | 0.75  |
| D     | 20.80    | 21.00 | 21.30 |
| D1    | 16.25    | 16.55 | 16.85 |
| E     | 15.50    | 15.80 | 16.10 |
| E1    | 13.00    | 13.30 | 13.60 |
| E2    | 4.8      | 5.00  | 5.20  |
| E3    | 2.3      | 2.50  | 2.70  |
| e     | 5.44 BSC |       |       |
| L     | 19.62    | 19.92 | 20.22 |
| L1    | -        | -     | 4.30  |
| Φ P   | 3.40     | 3.60  | 3.80  |
| Φ P1  | -        | -     | 7.30  |
| S     | 6.16 BSC |       |       |



**Revision History**

| Document Version | Date of Release | Description of Changes   |
|------------------|-----------------|--------------------------|
| Rev 0.0          | 2022-06-08      | Draft datasheet created. |
|                  |                 |                          |
|                  |                 |                          |
|                  |                 |                          |

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